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I. QUANTUM TOMOGRAPHY AT WORK

A. One Spin-1/2 system

Let's describe the one spin-1/2 density matrix:

$$\rho = \frac{1}{2}\left(\mathcal{I} + \vec{B}\cdot\vec{\sigma}\right) \tag{1}$$

If we want to analyze a decay $\tau/\overline{\tau} \to l^+/l^-$, then the option of meassure just two directions (up/down) isn't available, instead we need to meassure in a space, in a solid angle space. So, we have the intention of meassure in a specific direction in which the product of the decay goes on, say $\hat{n}$, then the proyector operator will take the form:

$$\Pi_{\hat{n}} = \frac{1}{2}\left(\mathcal{I} + \hat{n}\cdot\vec{\sigma}\right) \tag{2}$$

The probability that the decay product be emitted in the $\hat{n}$ direction is:

$$\begin{aligned}
\mathrm{p}(\hat{n}) &= \mathrm{Tr}\left[\frac{1}{2}\left(\mathcal{I} + \vec{B}\cdot\vec{\sigma}\right)\frac{1}{2}\left(\mathcal{I} + \hat{n}\cdot\vec{\sigma}\right)\right] \\
&= \mathrm{Tr}\left[\frac{1}{4}\left(\mathcal{I} + \vec{B}\cdot\vec{\sigma} + \hat{n}\cdot\vec{\sigma} + \underbrace{(\vec{B}\cdot\vec{\sigma})(\hat{n}\cdot\vec{\sigma})}_{(\vec{B}\cdot\hat{n})I + i(\vec{B}\times\hat{n})\cdot\vec{\sigma}}\right)\right] \\
&= \frac{1}{4}\mathrm{Tr}\left[\mathcal{I} + \vec{B}\cdot\vec{\sigma} + \hat{n}\cdot\vec{\sigma} + (\vec{B}\cdot\hat{n})I + i(\vec{B}\times\hat{n})\cdot\vec{\sigma}\right] \\
&= \frac{1}{4}\mathrm{Tr}\left[\mathcal{I} + \left(\vec{B} + \hat{n} + i(\vec{B}\times\hat{n})\right)\cdot\vec{\sigma} + (\vec{B}\cdot\hat{n})\mathcal{I}\right] \\
&= \frac{1}{4}\left[\underbrace{\mathrm{Tr}\left[\mathcal{I}\right]}_{4} + \underbrace{\mathrm{Tr}\left[\left(\vec{B} + \hat{n} + i(\vec{B}\times\hat{n})\right)\cdot\vec{\sigma}\right]}_{\vec{\sigma}\text{'s are traceless}} + \mathrm{Tr}\left[(\vec{B}\cdot\hat{n})I\right]\right] \\
&= \frac{1}{4}\left[2 + 2\vec{B}\cdot\hat{n}\right]
\end{aligned} \tag{3}$$

So, the probability is given by:

$$\mathrm{p}(\hat{n}) = \frac{1}{2}\left[1 + \vec{B}\cdot\hat{n}\right] \tag{4}$$

However, the probability isn't normalized in the solid angle space. Let's say that $\mathrm{P}(\hat{n}) \propto 1 + \vec{B}\cdot\hat{n}$, then:

$$\int d\Omega\ \mathrm{p}(\hat{n}) = 1 \tag{5}$$

$$\int d\Omega\left[c(1 + \vec{B}\cdot\hat{n})\right] = 1 \tag{6}$$

$$c\underbrace{\int d\Omega}_{4\pi} + c\vec{B}\cdot\underbrace{\int d\Omega\ \hat{n}}_{0} = 1 \tag{7}$$

$$c = \frac{1}{4\pi} \tag{8}$$

Finally, the probability density is given by:

$$\mathrm{p}(\hat{n}) = \frac{1}{4\pi}\left[1 + \vec{B}\cdot\hat{n}\right] \tag{9}$$

Finally, to perform Quantum Tomography we need to include a *Spin-analyzing power* parameter, κ, in order to cunatify how efficiently the $\hat{n}$ direction tracks the original polarization of the mother particle $\vec{B}$.

$$\mathrm{p}(\hat{n}) = \frac{1}{4\pi}\left[1+\kappa\vec{B}\cdot\hat{n}\right] \tag{10}$$

B. Important step to further probe

An important step that isn't detailled in the literature, and also I still don't completelly understand it, is that to obtain a full probability density function that the decay product be projected ($\hat{n}$) in a particular direction (+ or -), is multiply the probabilities of each path; that is:

$$\mathrm{p}(\hat{n}^+,\hat{n}^-) = \frac{1}{4\pi}\left[1+\kappa_+B_i^+n_i^+\right]\frac{1}{4\pi}\left[1+\kappa_-B_j^-n_j^-\right] \tag{11}$$

$$\mathrm{p}(\hat{n}^+,\hat{n}^-) = \frac{1}{(4\pi)^2}\left[1+\kappa_+B_i^+n_i^+ + \kappa_-B_j^-n_j^- + \kappa_+\kappa_-\sum_{i,j}C_{ij}n_i^+n_j^-\right] \tag{12}$$

C. Expected values

Since we already have an expression for the probability density function, the next step is try to find an expected value for a particular direction for the decay product.

1. $\langle\hat{n}^+\rangle$

$$\begin{aligned}
\langle n_i^+\rangle &= \int d\Omega^+ d\Omega^- n_l^+ \mathrm{p}(n^+,n^-) \\
&= \int d\Omega^+ d\Omega^- n_l^+ \frac{1}{(4\pi)^2}\left[1+\kappa_+B_i^+n_i^+ + \kappa_-B_j^-n_j^- + \kappa_+\kappa_-\sum_{i,j}C_{ij}n_i^+n_j^-\right] \\
&= \frac{1}{(4\pi)^2}\int d\Omega^+ d\Omega^-\left[n_l^+ + \kappa_+B_i^+n_l^+n_i^+ + \kappa_-B_j^-n_l^+n_j^- + \kappa_+\kappa_-\sum_{i,j}C_{ij}n_k^+n_i^+n_j^-\right] \\
&= \frac{1}{(4\pi)^2}\int d\Omega^+ d\Omega^-\left[\underbrace{n_l^+}_{(1)} + \underbrace{\kappa_+B_i^+n_l^+n_i^+}_{(2)} + \underbrace{\kappa_-B_j^-n_l^+n_j^-}_{(3)} + \underbrace{\kappa_+\kappa_-\sum_{i,j}C_{ij}n_l^+n_i^+n_j^-}_{(4)}\right]
\end{aligned} \tag{13}$$

$$\begin{aligned}
(1):&\quad \int d\Omega^+ d\Omega^- n_l^+ = \int \underbrace{d\Omega^-}_{4\pi}\int\underbrace{d\Omega^+ n_l^+}_{0} = 0 \\
(2):&\quad \int d\Omega^+ d\Omega^-\kappa_+B_i^+n_l^+n_i^+ = \kappa_+B_i^+\int\underbrace{d\Omega^-}_{4\pi}\int\underbrace{d\Omega^+n_l^+n_i^+}_{\frac{4\pi}{3}\delta_{li}} = \frac{(4\pi)^2}{3}\kappa_+B_l^+ \\
(3):&\quad \int d\Omega^+ d\Omega^-\kappa_-B_j^-n_l^+n_j^- = \kappa_-B_j^-\int\underbrace{d\Omega^+n_l^+}_{0}\int\underbrace{d\Omega^-n_j^-}_{0} = 0 \\
(4):&\quad \int d\Omega^+ d\Omega^-\kappa_+\kappa_-\sum_{i,j}C_{ij}n_l^+n_i^+n_j^- = \kappa_+\kappa_-\sum_{i,j}C_{ij}\int d\Omega^+n_l^+n_i^+\int\underbrace{d\Omega^-n_j^-}_{0} = 0
\end{aligned} \tag{14}$$

So, the expected value for the direction n_i^+ for the decay product is:

$$\langle n_i^+ \rangle = \frac{\kappa_+ B_i^+}{3} \tag{15}$$

or, generally:

$$\langle n_i^\pm \rangle = \frac{\kappa_\pm B_i^\pm}{3} \rightarrow B_i^\pm = \frac{3}{\kappa_\pm} \langle n_i^\pm \rangle \tag{16}$$

This $B_i^\pm$ coefficient it's a very usefull for the framework developed in this work. This is due to the contribution to find the density matrix of a two spin-1/2 system. Now, we still need to find an expression for the expected value for the projected direction of the decay product. Since we already have an expression to find it, there is another way to perform the calculation of this projected direction?

$$\langle n_i^+ \rangle = \int d\Omega^+ d\Omega^- n_i^+ \underbrace{\mathrm{p}(n^+, n^-)}_{\text{another expression?}} \tag{17}$$

2. *Find another $p(n^+, n^-)$*

In the particle physics context, the probability of produce a pair $\tau/\overline{\tau}$ and then decay to a product in the +/- direction came given by the cross section fo the proces. In this sense,

$$\int d\Omega^+ d\Omega^- \frac{d^2\sigma}{d\Omega^+ d\Omega^-} = \int d\sigma = \sigma_{total} \tag{18}$$

For a probability density function, this result should be normalized:

$$\int d\Omega^+ d\Omega^- \frac{1}{\sigma_{total}} \frac{d^2\sigma}{d\Omega^+ d\Omega^-} = 1 \rightarrow \mathrm{p}(n^+, n^-) \propto \frac{1}{\sigma} \frac{d^2\sigma}{d\Omega^+ d\Omega^-} \tag{19}$$

For instance, we can take this expression and perform some computation to get the expected value for a projected particular direction:

$$\begin{aligned} \langle n_i^+ \rangle &= \int d\Omega^+ d\Omega^- n_i^+ \frac{1}{\sigma} \frac{d^2\sigma}{d\Omega^+ d\Omega^-} \\ &= \frac{1}{\sigma} \int d\Omega^+ n_i^+ \frac{d\sigma}{d\Omega^+} \end{aligned} \tag{20}$$

or, generally,

$$\langle n_i^\pm \rangle = \frac{1}{\sigma} \int d\Omega^\pm n_i^\pm \frac{d\sigma}{d\Omega^\pm} \tag{21}$$

Let's replace this expression in [16]:

$$B_i^\pm = \frac{3}{\kappa_\pm} \frac{1}{\sigma} \int d\Omega^\pm n_i^\pm \frac{d\sigma}{d\Omega^\pm} \tag{22}$$

D. $\langle n_i^+ n_j^- \rangle$

Now, we will try to find another observable in this framework, something that can be meassure or can serve in order to obtain a complete description of the density matrix. Let's see the expected value of a product of projections:

$$\begin{aligned}
\langle n_i^+ n_j^- \rangle &= \int d\Omega^+ d\Omega^- n_i^+ n_j^- \mathrm{p}(n^+, n^-) \\
&= \int d\Omega^+ d\Omega^- n_i^+ n_j^- \frac{1}{(4\pi)^2} \left[1 + \kappa_+ B_l^+ n_l^+ + \kappa_- B_m^- n_m^- + \kappa_+ \kappa_- \sum_{l,m} C_{lm} n_l^+ n_m^- \right] \\
&= \frac{1}{(4\pi)^2} \int d\Omega^+ d\Omega^- \left[\underbrace{n_i^+ n_j^-}_{(1)} + \underbrace{\kappa_+ B_l^+ n_i^+ n_j^- n_l^+}_{(2)} + \underbrace{\kappa_- B_m^- n_i^+ n_j^- n_m^-}_{(3)} + \underbrace{\kappa_+ \kappa_- \sum_{l,m} C_{lm} n_i^+ n_j^- n_l^+ n_m^-}_{(4)} \right]
\end{aligned} \tag{23}$$

$$\begin{aligned}
(1): &\quad \int d\Omega^+ d\Omega^- n_i^+ n_j^- = \int \underbrace{d\Omega^- n_j^-}_{0} \int \underbrace{d\Omega^+ n_i^+}_{0} = 0 \\
(2): &\quad \int d\Omega^+ d\Omega^- \kappa_+ B_l^+ n_i^+ n_j^- n_l^+ = \kappa_+ B_l^+ \int d\Omega^+ n_i^+ n_l^+ \int \underbrace{d\Omega^- n_j^-}_{0} = 0 \\
(3): &\quad \int d\Omega^+ d\Omega^- \kappa_- B_m^- n_i^+ n_j^- n_m^- = \kappa_- B_m^- \int \underbrace{d\Omega^+ n_i^+}_{0} \int d\Omega^- n_j^- n_m^- = 0 \\
(4): &\quad \int d\Omega^+ d\Omega^- \kappa_+ \kappa_- \sum_{l,m} C_{lm} n_i^+ n_j^- n_l^+ n_m^- = \kappa_+ \kappa_- \sum_{l,m} C_{lm} \int \underbrace{d\Omega^+ n_i^+ n_l^+}_{} \frac{4\pi}{3} \delta_{il} \int \underbrace{d\Omega^- n_j^- n_m^-}_{\frac{4\pi}{3}\delta_{jm}} \\
&= \left(\frac{4\pi}{3} \right)^2 \kappa_+ \kappa_- \sum_{l,m} C_{lm} \delta_{jm} \delta_{il} = \left(\frac{4\pi}{3} \right)^2 \kappa_+ \kappa_- C_{ij}
\end{aligned} \tag{24}$$

So,

$$\langle n_i^+ n_j^- \rangle = \frac{\kappa_+ \kappa_-}{9} C_{ij} \tag{25}$$

$$C_{ij} = \frac{9}{\kappa_+ \kappa_-} \langle n_i^+ n_j^- \rangle \tag{26}$$

Taking acount of Eq. [19], we have:

$$C_{ij} = \frac{9}{\kappa_+ \kappa_-} \frac{1}{\sigma} \int d\Omega^+ d\Omega^- n_i^+ n_j^- \frac{d^2\sigma}{d\Omega^+ d\Omega^-} \tag{27}$$

E. Probabilities

We have several ways to calculate the probabilities that the product decay came in a particular direction. This correlation can also appear in a way of probabilities. Let's recall Eq. [10] y Eq. [19]; the probabilities for the same case should be the same, as:

$$\begin{aligned}
\frac{1}{4\pi} \left[1 + \kappa \vec{B} \cdot \hat{n} \right] &= \frac{1}{\sigma} \frac{d^2\sigma}{d\Omega^+ d\Omega^-} \\
\frac{1}{4\pi} \left[1 + \kappa \vec{B}_i^+ \hat{n}_i^+ \right] &= \frac{1}{\sigma} \frac{d\sigma}{d\Omega^+} \longrightarrow \text{Only the } ^+ \text{ direction}
\end{aligned} \tag{28}$$

Because we're looking a particular direction of decay in a solid angle, the following equations are necessary for our task:

$$\vec{B}^{+}\cdot\hat{n}^{+} = B_x^{+}\sin\theta_i^{+}\cos\phi_i^{+} + B_y^{+}\sin\theta_i^{+}\sin\phi_i^{+} + B_z^{+}\cos\theta_i^{+} \tag{29}$$

$$d\Omega^{\pm} = d\cos\theta^{\pm}d\phi^{\pm} \tag{30}$$

Take the Eq. [28] and integrate over the azimutal angle:

$$\begin{gathered}
\frac{1}{4\pi}\left[1+\kappa\vec{B}\cdot\hat{n}\right] = \frac{1}{\sigma}\frac{d\sigma}{d\Omega^{+}} \quad \cdot\int d\phi^{+} \\
\frac{1}{4\pi}\underbrace{\int_0^{2\pi}d\phi^{+}}_{2\pi} +\kappa_{+}B_x^{+}\sin\theta_i^{+}\underbrace{\int_0^{2\pi}d\phi^{+}\cos\phi_i^{+}}_{0} +B_y^{+}\sin\theta_i^{+}\underbrace{\int_0^{2\pi}d\phi^{+}\sin\phi_i^{+}}_{0} +B_z^{+}\cos\theta_i^{+}\underbrace{\int_0^{2\pi}d\phi^{+}}_{2\pi} = \frac{1}{\sigma}\int_0^{2\pi}d\phi^{+}\frac{d\sigma}{d\Omega^{+}} \\
\frac{1}{4\pi}\left[2\pi+2\pi B_z^{+}\cos\theta_i^{+}\right] = \frac{1}{\sigma}\int_0^{2\pi}d\phi^{+}\frac{d\sigma}{d\cos\theta^{+}d\phi^{+}} \\
\frac{1}{2}\left[1+B_i^{+}\cos\theta_i^{+}\right] = \frac{1}{\sigma}\frac{d\sigma}{d\cos\theta^{+}}
\end{gathered} \tag{31}$$

Because the only surviving term is B_z, we choose a quantization axis in which $\hat{n}\cdot\hat{z} = \cos\theta_i$. In this framework, the remaining value of κ depends on the process of interest; in this direction, $\kappa = \pm 1$, so the final result is as following:

$$\frac{1}{\sigma}\frac{d\sigma}{d\cos\theta_i^{+}} = \frac{1}{2}\left(1+B_i^{+}\cos\theta_i^{+}\right) \tag{32}$$

If now, we focus on both directions of the decay products, we need to work in both solid angle spaces:

$$\begin{gathered}
\frac{1}{\sigma}\frac{d^2\sigma}{d\Omega^{+}d\Omega^{-}} = \frac{1}{(4\pi)^2}\left[1+\kappa_{+}\vec{B}^{+}\cdot\hat{n}^{+}\right]\left[1+\kappa_{-}\vec{B}^{-}\cdot\hat{n}^{-}\right] \quad \cdot\int d\phi^{+}d\phi^{-} \\
\frac{1}{\sigma}\int d\phi^{+}d\phi^{-}\frac{d^2\sigma}{d\cos\theta_i^{+}d\phi_i^{+}d\cos\theta_j^{-}d\phi_j^{-}} = \frac{1}{(4\pi)^2}\int_0^{2\pi}d\phi^{+}d\phi^{-}\left[1+\kappa_{+}B_i^{+}n_i^{+}+\kappa_{-}B_j^{-}n_j^{-}+\kappa_{+}\kappa_{-}B_i^{+}B_j^{-}n_i^{+}n_j^{-}\right] \\
\frac{1}{\sigma}\frac{d^2\sigma}{d\cos\theta_i^{+}d\cos\theta_j^{-}} = \frac{1}{(4\pi)^2}\left[(2\pi)^2+\right]
\end{gathered} \tag{33}$$